

Chapter 6, Problem 7RQ

Problem

A 5-H inductor changes its current by 3 A in 0.2 s. The voltage produced at the terminals of the inductor is:

- (a) 75 V
- (b) 8.888 V
- (c) 3 V
- (d) 1.2 V

Step-by-step solution

Step 1 of 2

The expression for the voltage across an inductor is,

$$v = L \frac{di}{dt}$$

di is the change in current.

dt is the change in time.

The current through the inductor cannot undergo an abrupt change, as it needs an infinite amount of voltage.

So, the current must change a finite amount in a finite time period.

Step 2 of 2

Recall the expression for the voltage across the inductor.

$$v = L \frac{di}{dt}$$

Substitute 5-H for L , 3A for di and 0.2 s for dt .

$$\begin{aligned} v &= 5 \times \frac{3}{0.2} \\ &= 5 \times 15 \\ &= 75 \text{ V} \end{aligned}$$

Thus, the voltage produced across the terminals of the inductor is 75 V.